

User Guide: CSV to Estimate Output

Distribution Analysis & Navigation Algorithm (D.A.N.A.) Framework

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Component:	CSV to Estimate Output Guide	Input File:	Shift Impact CSV

Welcome to the CSV to Estimate Output guide. This guide provides structured steps for library project managers to seamlessly transition raw computational shift files into actionable, on-the-ground shelving estimates.

1. Importing Data into the Target Workbook

To automatically format the downloaded `Shift_Impact_Ranges_xx_to_xx.csv` file from the Library Space Management Tool, follow these exact data placement rules:

1. Locate and open the pre-formatted workbook called “Formatted Estimate From Rawdata CSV” located in the Distribution Analysis & Navigation Algorithm Framework folder.
2. Open your downloaded `Shift_Impact_Ranges_xx_to_xx.csv` file using Excel or Google Sheets.
3. Copy all of the active data held within that CSV spreadsheet.
4. Return to the destination workbook, select the sheet tab labeled “**Rawdata**”, and paste the copied data.
5. Click on the “**Dashboard**” sheet in the “**Formatted Estimate From Rawdata CSV**” workbook to view the updated estimate guide.

CRITICAL DATA ALIGNMENT STEP

In your formatted destination workbook, you must copy and paste the corresponding “**Average Shelf Length**” values to their correct range sides used for the shift. This data was entered in your original Data Collection Workbook column ‘H’.

2. Navigating Shifting Estimates and Buffers

The dashboard translates raw mathematical linear numbers into physical operations on the library floor.

- **The “Moved In” and “Moved Out” Columns:** Highlight the precise physical areas that the D.A.N.A. program allotted to be moved in order to satisfy your shifting targets.

2.1 Movable Capacity (Shift Target)

What it means: The exact number of empty shelves available to absorb moved materials. Note this number will not match the total of empty shelves counted after each range.

How to use it: This is the number you plan against. If this number is equal to or less than zero there will not be enough space to continue shifting materials. When a negative value appears in your ‘Movable Capacity’ column, this signals that you may need to “backshift” (move collections in reverse order). To accomplish this, you must adjust the functional formula in the ‘Movable Capacity’ column.

2.2 Physical Empty Shelves (Accuracy Check)

What it means: The total number of physically empty shelves you will count after shifting a range.

Why it is higher than Movable Capacity: It includes both your Movable Capacity plus the 'Target Available'. This is the amount of space intentionally left per shelf designated by the value you set for your 'Target Empty Percentage'.

How to use it: Use this number when auditing shelves per range after shifting materials. This is used to confirm your physical counts match the spreadsheet.

2.3 Target Available (Percentage of Empty Space Per Shelf)

What it means: The required empty space permanently set aside in each range (e.g., leaving 25% of each shelf empty for collection growth post shift).

Rule of Thumb: Do not fill these shelves. They are included in your physical post range shift shelf count, but they are protected from receiving shifted material. Physical Empty Shelves is used to check accuracy but Movable Capacity indicates a safe achievable shift.

Simplified: Physical Empty Shelves is how many shelves you will count after shifting a range. It is the easiest way to know if you are on target.

FORMULA EXPLANATION

The formula $=I4+(D4/H4)$ calculates the total physical empty shelves you will count on the floor by combining two parts:

D4/H4 (Reserved Buffer in Shelves): Converts your target expansion space in inches (D4) into whole shelf units by dividing it by the average shelf length (H4).

I4 + ... (Total Physical Count): Adds those reserved buffer shelves back to your net movable space estimate (I4).

In short: It takes the space you are allowed to move material into (I4) and adds the extra empty space held back for growth (D4/H4), giving you the true number of physically empty shelves sitting in the aisle.

HANDLING REAL-WORLD SHELF LOGISTICS

Because it is virtually impossible during physical shifts to isolate and relocate only the exact linear space required, the guide assumes every shelf will be shifted to your chosen target percentage. It leaves it to the project lead to decide where to leave surplus space behind.

Buffer Recommendation: We highly recommend building in an extra **3%–5% buffer** on top of the estimated physical move targets. Any surplus space accumulated beyond this 5% buffer is advised to be left behind as the shift progresses either at the end of a range side, or strategically distributed throughout the range side itself.

3. Advanced Troubleshooting: Manual Backshifting Equations

If negative values appear in your **'Movable Capacity'** column, this signals that you will need to "backshift" (move collections in reverse order). To accomplish this, you must adjust the functional formula in the 'Movable Capacity' column.

3.1 Demystifying the Formula Logic

It is helpful to examine the logical differences between the first and second cells in the **'Movable Capacity'** column to apply custom manual tweaks safely:

- **First Cell Equation:** `=iferror(-(E4/H4))`
- **Second Cell Equation:** `=iferror(I4-(E5/H5))`

How the Equations Function:

- The `iferror` wrapping is strictly implemented to prevent the spreadsheet from filling with error messages (such as `#DIV/0!`) when no data has been entered in the **"Avg. Shelf Length"** column.
- **The first equation:** Divides the 'Initial Need/Surplus' by the 'Avg. Shelf Length'. This result is then multiplied by a negative 1 (`-`) so that the 'Movable Capacity' column displays a positive number, illustrating exactly how many empty shelves will remain after shifting a range side.
- **The second equation:** Follows a very similar logic, except it dynamically adds the remaining shelves from the previous cell in the 'Movable Capacity' column after the current row's 'Initial Need/Surplus' is divided by 'Avg. Shelf Length'. Once this second cell formula is set, it can be dragged down/up to copy the formula through your remaining rows.

4. Helpful Tips for Project Leads

The workbook is designed to systematically track surplus space throughout the move. It is the responsibility of the project lead to decide what to do with extra capacity beyond what is needed to complete the shift. This transparency removes the traditional guesswork, giving you full awareness of whether you are collecting too many or too few surplus shelves.

SIMPLE TIPS FOR SUCCESS

Tip 1 (Formula Check): Ensure that the 'Movable Capacity' column does not require manual backshifting adjustments.

Tip 2 (Launch the Shift): Once formula errors and backshifts are ruled out, start the physical shift following your chosen target percentage.

Tip 3 (Accumulate Buffer): If you accumulate surplus shelves, let them build up until you have an amount that is 3%–5% over what is physically needed to complete the shift.

Tip 4 (Store Surplus): After this threshold occurs, always leave any additional surplus space behind as you continue moving forward.

Tip 5 (Final Reduction): As the shift nears completion, start reducing and utilizing the 3–5% surplus buffer if it still remains.

Tip 6 (Shifting Stick): Create a shifting "stick" that will be the precise size of the Target Empty Percentage chosen by the project lead. Remember this will be the space from the end of a shelf to the physical book.

Tip 7 (Shifting Clarity): For a full understanding of your shifting efficiency perform a post shift measurement.